Data-Driven Analysis of Product, Regional, Shipping, and Factory-Reference Performance at Nassau Candy Distributor

A Data Analytics, Predictive Modelling, Clustering, and Decision-Support Study

Abstract

This research presents a data-driven analysis of product, regional, shipping, and factory-reference performance at Nassau Candy Distributor. The study uses a dataset containing 10,194 records and 18 original variables to examine business patterns, assess data quality, engineer analytical features, and develop decision-support insights.

The analytical methodology includes exploratory data analysis, lead-time predictive modelling, clustering, factory-reference analysis, scenario simulation, profitability sensitivity analysis, optimization recommendations, and KPI and risk assessment. A Streamlit dashboard was also developed to present key findings, interactive analyses, and decision-support information.

The study aims to support a better understanding of product performance, regional patterns, shipping characteristics, and factory-reference options.

It also demonstrates how predictive analytics and analytical tools can support business review and planning. However, factory-reference mappings are treated as reference information rather than verified historical dispatch records, and scenario outputs are interpreted within the limits of available evidence.

The findings provide an analytical foundation for business review and future investigation. The study also identifies data and methodological limitations that should be considered before operational decisions are made.

Introduction

The distribution industry involves several interconnected business activities, including product management, order processing, shipping, regional distribution, and supply chain coordination. Analysing these activities is important for

understanding business performance and identifying opportunities for improvement.

Nassau Candy Distributor provides the business context for this research. The project uses historical business data to examine product performance, regional patterns, shipping characteristics, and factory-reference information.

Traditional reporting methods can describe past business performance, but analytical techniques can provide additional insights. Exploratory data analysis helps identify patterns and relationships, predictive modelling supports the examination of lead-time behaviour, and clustering helps identify groups of similar business records. Scenario simulation and optimization methods can further support structured business evaluation.

This research applies a combination of data analytics, machine learning, statistical analysis, and decision-support techniques to the available Nassau Candy Distributor dataset. It also presents the results through an interactive Streamlit dashboard.

The overall purpose is to develop a structured analytical framework that helps interpret business data, evaluate operational patterns, and support informed management review. The research also considers the limitations of the available data to ensure that analytical findings are not interpreted as unsupported operational guarantees.

Business Background

Nassau Candy Distributor is the business context for this research project.

The distribution business involves managing products, customer orders, shipment activities, and the movement of goods across different regions.

Effective distribution management requires an understanding of product demand, sales performance, order characteristics, shipping methods, and delivery-related patterns. Analysing these areas can help identify differences in business performance and provide useful information for management review.

This project examines historical transaction-level data associated with

Nassau Candy Distributor. The analysis focuses on product performance, regional performance, shipping performance, factory-reference analysis, and business decision support.

The project combines descriptive analytics, predictive modelling, clustering, scenario simulation, profitability sensitivity analysis, and optimization-oriented methods. These techniques are presented through an interactive Streamlit dashboard to make the analytical results easier to explore.

An important consideration is that the available dataset does not establish every operational detail. Factory-reference mappings should not be interpreted as verified historical dispatch records, and alternative factory scenarios should not be treated as confirmed cost savings or operational improvements. The analysis therefore distinguishes observed data from assumptions and areas requiring further validation.

Problem Statement

The availability of historical business data creates opportunities to understand product performance, regional differences, shipping characteristics, and operational patterns. However, raw transaction data alone does not automatically provide clear and actionable business insights.

The main problem addressed by this research is the need to transform the available Nassau Candy Distributor dataset into a structured analytical framework that can support business performance evaluation and informed decision-making.

The research examines product-level sales and profitability, regional performance, shipping methods, order lead-time behaviour, predictive modelling, clustering, factory-reference mappings, scenario simulation, profitability sensitivity, optimization recommendations, and business risks.

A further challenge is ensuring that analytical conclusions remain consistent with the available evidence. Reference factory mappings do not establish actual

historical dispatch decisions, and scenario simulations cannot independently confirm factory-specific cost savings or operational improvements.

Therefore, the research focuses on developing reliable analytical insights while clearly distinguishing measured results, model-based estimates, reference information, and assumptions requiring additional validation.

Research Objectives

The primary objective of this research is to develop a data-driven analytical framework for examining product, regional, shipping, and factory-reference performance at Nassau Candy Distributor.

The specific objectives are:

1. To understand the structure, characteristics, and quality of the available business dataset.
2. To perform exploratory data analysis to examine product, regional, shipping, sales, and profitability patterns.
3. To engineer relevant analytical features from the original business data.
4. To develop and evaluate a predictive model for analysing order lead-time behaviour.
5. To apply clustering techniques to identify groups of similar business observations.
6. To examine product-to-factory reference mappings and evaluate alternative factory options using the available evidence.
7. To conduct scenario simulation and profitability sensitivity analysis.
8. To develop optimization-oriented recommendations and assess relevant business KPIs and risks.
9. To create an interactive Streamlit dashboard for communicating analytical results and supporting business review.
10. To identify limitations and opportunities for future research.

Research Questions

This research is guided by the following questions:

1. What patterns can be identified in product-level sales and profitability?

2. How does business performance vary across geographical regions?

3. What patterns are visible across shipping methods and order lead times?

4. Can historical business features support predictive modelling of order lead-time behaviour?

5. Can clustering identify meaningful groups within the available business observations?

6. What insights can be obtained from product-to-factory reference mappings?

7. How can alternative factory scenarios be assessed when factory-specific evidence is incomplete?

8. How can profitability sensitivity and optimization-oriented analysis support business review?

9. How can an interactive dashboard improve access to analytical findings?

10. What data limitations should be considered before operational decisions are made?

Dataset Description

The research uses the Nassau Candy Distributor dataset supplied for this project.

The original dataset contains 10,194 records and 18 columns. The processed analytical dataset contains 10,194 records and 34 columns after the project's data preparation and feature engineering activities.

The data includes business information used to examine products, orders, sales, profitability, geographical regions, shipping characteristics, and related analytical variables.

The original dataset is retained separately from the processed dataset to preserve data traceability. Date fields are interpreted using day-first

parsing where required, without unnecessarily overwriting the original date strings.

The processed dataset supports exploratory analysis, predictive modelling, clustering, scenario evaluation, and dashboard reporting.

The predictive model uses 20 predictor features. Factory information is not included as a predictor in that model.

The dataset represents the records available for this study. It should not automatically be interpreted as a complete representation of every operational event or supply-chain decision made by the business.

Data Quality Methodology

Data quality assessment was conducted before the main analytical activities. The purpose was to identify issues that could affect the reliability and interpretation of the results.

The assessment included:

- Inspection of dataset dimensions and column names.
- Review of data types and expected variable formats.
- Identification of missing values.
- Examination of duplicate records.
- Validation of date fields and date parsing.
- Review of numeric variables and unusual values.
- Examination of categorical values and inconsistent labels.
- Validation of important derived variables.

Original data was preserved to support traceability. Transformations were performed in the processed dataset rather than silently changing the raw source.

Date parsing followed the day-first format where appropriate. Lead time was calculated as the difference between Ship Date and Order Date. Negative or otherwise invalid calculated values were considered during validation.

Lead time in this research represents the interval between the recorded order date and ship date. It is not an independently verified measure of carrier transit time.

Data quality checks were used to improve consistency and reduce the risk of misleading analytical conclusions. Remaining limitations were documented rather than hidden.

Feature Engineering Methodology

Feature engineering was used to transform the original business variables into features suitable for descriptive and predictive analysis.

The process included date-related feature preparation, financial feature calculations, safe numerical operations, and model feature preparation.

Order-date features were derived to support the examination of temporal patterns. Financial features were prepared to support revenue and profitability analysis. Safe division methods were used where ratios or other division-based measures were required, helping avoid invalid calculations caused by zero denominators.

Lead time was calculated using the recorded Ship Date and Order Date. Additional lead-time categories were used for grouped analysis.

The model preparation process selected 20 predictor features for the lead-time prediction pipeline. Factory was not included as a model predictor.

Feature engineering was performed with attention to data types, missing values, and consistency between training and prediction workflows.

Derived features are analytical representations of the available source data. They should not be interpreted as independently verified operational measurements.

Exploratory Data Analysis Results

Exploratory data analysis was conducted to understand the characteristics and patterns of the available business data.

The analysis examined product performance, regional performance, shipping methods, sales, profitability, and lead-time behaviour.

Product-level analysis was used to compare business performance across products. Regional analysis examined geographical differences in the available records. Shipping analysis compared order characteristics and lead-time patterns across shipping methods.

Financial analysis provided a basis for reviewing sales and profitability indicators. Visualizations and summary statistics were used to make patterns easier to interpret.

The exploratory analysis provides the descriptive foundation for the subsequent predictive, clustering, scenario, and decision-support activities.

The final submitted paper should include the verified numerical findings and relevant charts from the exploratory analysis notebook. In particular, the following values should be inserted from the actual outputs:

- Leading products by the selected sales or profitability measure:

[INSERT VERIFIED PRODUCT FINDINGS]

- Regional performance observations:

[INSERT VERIFIED REGIONAL FINDINGS]

- Shipping method comparisons:

[INSERT VERIFIED SHIPPING FINDINGS]

- Overall sales and profitability observations:

[INSERT VERIFIED FINANCIAL FINDINGS]

- Lead-time distribution and notable patterns:

[INSERT VERIFIED LEAD-TIME FINDINGS]

No numerical ranking or business conclusion should be added unless it is supported by the completed analysis.

Predictive Modelling Methodology

Predictive modelling was used to examine lead-time behaviour using the available historical business features.

The modelling workflow included preparing the target variable, selecting predictor features, preparing the input data, defining a machine learning pipeline, training the model, and saving the resulting model artifact.

The model uses 20 predictor features. Factory information is not included as a predictor.

A pipeline-based approach was used to maintain consistency between model preparation and prediction. The saved model artifact was subsequently reloaded to verify that it could be used outside the original training session.

The modelling process was designed to support reproducibility and consistent prediction behaviour. Repeated predictions and predictions after reloading the saved pipeline were compared during validation.

The model estimates lead-time behaviour based on the features and relationships represented in the available dataset. Its predictions should be interpreted as model estimates rather than guaranteed future delivery outcomes.

Model Evaluation

Model evaluation was performed to assess the predictive performance and reliability of the lead-time modelling workflow.

Evaluation should be based on the actual metrics generated by the completed modelling notebook. Relevant regression measures may include Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R-squared), where these were calculated in the project.

The model artifact was reloaded successfully as a machine learning Pipeline. Prediction reproducibility was also checked by comparing repeated predictions and predictions made after reloading the saved artifact.

The following verified results must be inserted before final submission:

- Model algorithm: [INSERT ACTUAL MODEL ALGORITHM]
- MAE: [INSERT VERIFIED VALUE]
- MSE: [INSERT VERIFIED VALUE, IF CALCULATED]
- RMSE: [INSERT VERIFIED VALUE]
- R-squared: [INSERT VERIFIED VALUE]
- Test dataset size: [INSERT VERIFIED TEST SIZE]

If multiple model evaluation outputs differ, the discrepancy should be documented and investigated. Values must not be silently reconciled or replaced with estimates.

Evaluation metrics describe performance under the project's evaluation conditions. They do not guarantee the same accuracy on future or substantially different business data.

Clustering Methodology

Clustering analysis was performed to identify groups of similar business observations based on selected analytical features.

The purpose of clustering was to explore patterns that may not be immediately visible through individual variables or conventional summary statistics.

The workflow involved selecting relevant variables, preparing the data, applying the clustering approach used in the project, and examining the resulting group characteristics.

Cluster profiles were reviewed to understand how groups differed across the selected business dimensions. Appropriate visualizations and summary

statistics were used to support interpretation.

The resulting clusters represent similarities within the selected data and features. They should not automatically be interpreted as official business segments, customer types, or operational categories without additional business validation.

The final paper should report the actual clustering algorithm, selected features, number of clusters, and verified cluster characteristics from the completed clustering notebook.

Factory Reference Analysis

Factory-reference analysis examined the relationship between products and their associated factory reference information.

The purpose was to understand the available product-to-factory mapping and review alternative factory options within the limits of the data.

The analysis distinguishes the reference factory mapping from verified historical dispatch information. A product-to-factory mapping indicates a reference relationship in the available data; it does not independently prove that a particular factory fulfilled a specific historical order.

Alternative factory options were reviewed using available compatibility and recommendation information. These options require additional operational validation before they can be treated as feasible production or distribution alternatives.

The analysis therefore focuses on organizing available reference information and identifying options that may warrant further investigation. It does not establish actual factory capacity, confirmed dispatch history, or guaranteed operational improvements.

Scenario Simulation

Scenario simulation was used to examine alternative analytical

conditions and support structured business review.

The simulation process considered available factory-option information and relevant supporting evidence. The project included checks for factory-specific lead-time estimability, factory-specific cost availability, and capacity data availability.

A scenario was treated as requiring further validation when the necessary factory-specific evidence was available. Where the evidence was incomplete, the scenario was classified as having insufficient factory-specific support.

The scenario assessment deliberately avoids inventing factory-specific lead-time changes, cost savings, or probabilities of success when these cannot be supported by the available information.

Scenario outputs should therefore be interpreted as structured assessments of available evidence, not as confirmed forecasts or operational guarantees.

Optimization Methodology

Optimization-oriented analysis was conducted to organize business priorities and support evidence-based review.

The methodology used available performance indicators and recommendation scores to rank or prioritize existing business observations for further investigation.

Existing product review priorities were interpreted as analytical priorities rather than proof of factory benefit, operational feasibility, or guaranteed financial improvement.

The project also examined factory-option information and supporting evidence to distinguish options that may require additional validation from options for which the available evidence is insufficient.

The recommendations are indicative and depend on the quality and

completeness of the available dataset. They should be reviewed alongside business constraints, operational knowledge, capacity information, and verified cost data before implementation.

The analysis does not claim to identify a mathematically proven global optimum or a guaranteed best factory allocation.

KPI and Risk Analysis

KPI and risk analysis was conducted to organize important business indicators and highlight areas requiring management attention.

The analysis considered available indicators associated with product performance, regional performance, shipping, lead time, profitability, and factory-reference evaluation.

Risk assessment focused on interpreting analytical findings within the limits of the available evidence. Particular attention was given to data quality, model uncertainty, factory-specific evidence gaps, and the distinction between reference mappings and actual operational records.

The project recognizes that a predictive estimate is not a guaranteed business outcome. Similarly, scenario support does not establish a probability of success, and an indicative recommendation does not prove operational feasibility.

The KPI and risk analysis is intended to support monitoring and structured review. Final KPI definitions, thresholds, and management actions should be aligned with verified business requirements.

Dashboard Description

An interactive Streamlit dashboard was developed to communicate the project's analytical findings and provide a practical interface for exploring the available data.

The dashboard contains eight main pages:

1. Home
2. Product Performance
3. Region Analysis
4. Ship Mode Analysis
5. Factory Explorer
6. Scenario Simulator
7. Recommendations
8. Risk and Impact

The application includes relevant filters for product, region, and shipping method. It also presents KPI cards, charts, scenario controls, recommendation panels, and risk information.

CSV and Excel export functionality is provided to support further review of the analytical data.

The application uses reusable utility modules for prediction, feature preparation, scenario assessment, and recommendation logic.

Validation checks were performed on the application source, model loading, filters, scenario logic, and data export functionality.

The dashboard is a decision-support interface. It does not replace business validation, operational planning, or management approval.

Results and Discussion

The project combines descriptive analytics, predictive modelling, clustering, factory-reference analysis, scenario simulation, profitability sensitivity analysis, optimization-oriented recommendations, and KPI and risk assessment.

The exploratory analysis provides an overview of product, regional, financial, and shipping patterns in the available records. Feature engineering prepares the dataset for further analytical activities.

The lead-time modelling workflow demonstrates how historical business features can be used to estimate lead-time behaviour. Model reproducibility and artifact reload checks provide additional evidence that the modelling workflow can be reused consistently.

Clustering provides a way to examine similarities between business observations. Factory-reference analysis organizes available mapping information, while scenario simulation distinguishes supported assessments from options that require additional evidence.

The recommendation and risk analysis translates analytical outputs into structured areas for business review. The Streamlit dashboard makes these outputs accessible through interactive pages and downloadable reports.

The overall contribution is an integrated analytical workflow that connects data preparation, analysis, modelling, and decision support.

The final discussion should incorporate the verified numerical results, model evaluation metrics, cluster profiles, and most important findings from the completed project notebooks. Any differences between evaluation outputs should be disclosed rather than concealed.

Research Limitations

This research has several limitations that should be considered when interpreting its findings.

First, the analysis is limited to the information contained in the available dataset. Missing, incomplete, or unrecorded business information may affect the interpretation of results.

Second, calculated lead time is based on the difference between the recorded Ship Date and Order Date. It is not an independently verified measure of carrier transit time.

Third, the product-to-factory relationship is treated as reference mapping information. It does not establish verified historical dispatch or fulfilment decisions.

Fourth, factory-specific capacity, cost, and lead-time evidence may not be available for every alternative factory option. Consequently, scenario assessments cannot establish confirmed factory-specific savings or operational improvements.

Fifth, Haversine distances represent straight-line geographical distances. They do not represent actual road distances, freight routes, or transportation costs.

Sixth, model performance depends on the available data, selected features, and evaluation methodology. Predictive results should not be treated as guaranteed future outcomes.

Seventh, clustering results depend on the selected features and clustering methodology. Identified groups require business interpretation and validation.

Finally, optimization-oriented recommendations are indicative priorities rather than proven globally optimal decisions. Additional operational and financial information is needed before implementation.

Future Scope

Future research can extend this project by incorporating additional operational and business information.

Potential improvements include:

- Integrating verified historical factory dispatch and fulfilment records.
- Incorporating factory-specific production capacity and availability.
- Collecting verified manufacturing and transportation cost data.
- Including actual road and freight-route distances instead of relying

only on straight-line geographical distances.

- Incorporating carrier transit events and delivery confirmation dates.
- Testing additional predictive models and evaluating them using consistent validation procedures.
- Monitoring model performance as new business data becomes available.
- Investigating additional clustering methods and validating cluster interpretations with business stakeholders.
- Developing more comprehensive scenario simulations using verified operational constraints.
- Extending optimization analysis with validated cost, capacity, compatibility, and service-level constraints.
- Establishing formal KPI definitions and monitoring thresholds.
- Improving dashboard deployment, access control, and automated data refresh processes.

These improvements could increase the practical value of the analytical framework and support more detailed operational decision-making.

Conclusion

This research developed a data-driven analytical framework for examining product, regional, shipping, and factory-reference performance at Nassau Candy Distributor.

Using a dataset containing 10,194 records and 18 original columns, the project carried out data quality assessment, feature engineering, exploratory data analysis, lead-time predictive modelling, clustering, factory-reference analysis, scenario simulation, profitability sensitivity analysis, optimization-oriented recommendations, and KPI and risk assessment.

An interactive Streamlit dashboard was also developed to present analytical findings through eight main pages, filters, visualizations,

scenario controls, and downloadable reports.

The project demonstrates how different analytical techniques can be combined into a structured decision-support workflow. It also emphasizes the importance of interpreting results according to the evidence available.

In particular, reference factory mappings should not be treated as verified dispatch records, scenario assessments should not be interpreted as guaranteed cost savings, and predictive estimates should not be considered certain future outcomes.

Overall, the research provides an analytical foundation for business review and identifies areas where additional operational data and validation could strengthen future decision-making.

References

The following references provide methodological background for the analytical techniques and software used in this project.

The exact software versions and project-specific sources should be verified before final submission.

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4. Hunter, J. D. (2007).
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5. The pandas development team.

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6. The scikit-learn developers.

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7. Streamlit, Inc.

Streamlit documentation.

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8. Nassau Candy Distributor dataset.

Original project dataset supplied for this research.

Add the appropriate source, access details, or institutional

attribution required by the submission guidelines.

Only include references that were actually consulted and are

relevant to the final submitted research paper.

Final Formatting and Review

Before submitting the research paper, complete the following review:

[ ] Confirm the final title and subtitle.

[ ] Review the abstract for accuracy and completeness.

[ ] Check the introduction and business background.

[ ] Confirm that the problem statement and research questions

match the completed project.

[ ] Verify dataset dimensions and variable descriptions.

[ ] Add actual data quality findings.

[ ] Verify feature engineering descriptions.

[ ] Insert numerical EDA findings and relevant charts.

[ ] Confirm the actual predictive model algorithm.

[ ] Insert verified model evaluation metrics.

[ ] Add actual clustering methodology and cluster findings.

[ ] Verify factory-reference analysis descriptions.

[ ] Confirm scenario simulation evidence limitations.

[ ] Review optimization recommendations.

[ ] Add relevant KPI and risk findings.

[ ] Include dashboard screenshots if required.

[ ] Verify all figures, tables, captions, and numbering.

[ ] Check that every reference was actually consulted.

[ ] Check spelling, grammar, and academic formatting.

[ ] Confirm that no unsupported savings, predictions, or operational claims are presented as established facts.

[ ] Save the final paper in the required submission format.

The research paper should follow the formatting and citation requirements specified by the relevant institution or evaluator.